1—Geometric constructions

A 5 d They meet at a common point.

3—Constructing quadrilaterals

A 1 a rectangle b parallelogram c rhombus
d square e trapezium f rhombus
g kite h trapezium

C 1 c $AB = 45$ mm, $BC = 28$ mm, $CD = 28$ mm, $AD = 45$ mm
d kite
2 c All sides measure 36 mm. d rhombus

4—Constructions and circles

A 2 a yes b no c no d no

C 3 a 72° b 54° 5 a 45°

5—Similar figures

A 1 *A–B, C–H, D–N, E–P, F–L, G–I, J–Q, K–O, M–R*
2 a The length of rectangle B is *two* times the length of rectangle A. The breadth of rectangle B is *two* times the length of rectangle A. b 1 : 2
3 a yes b true

B A : H = 1 : 2, B : F = 5 : 4, C : I = 5 : 8, D : E = 1 : 2, G : J = 5 : 4

C 1 8 cm 2 9 cm 3 6 cm 4 18 cm 5 9 cm
6 12 cm

7—Similar triangles

A 1 $\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$
3a 0.5 3b 0.5 3c 0.5 4 1 : 2

B 1 yes 2 no 3 yes
4 A–L, B–N, C–K, D–O, E–P, F–Q, G–R, H–M

C 1 In the Δs ABC and ADE,
$\angle BAC$ is common
$\angle ABC = \angle ADE$
$\angle ACB = \angle AED$ (equal corresponding angles, $BC \parallel DE$)
$\therefore \Delta ABC \,|||\, \Delta ADE$ (They are equi-angular.)

2 In the Δs PRT and QST,

$$\frac{PT}{QT} = \frac{6}{9} = \frac{2}{3} \text{ and } \frac{RT}{ST} = \frac{4}{6} = \frac{2}{3}$$

$\angle PTR = \angle QTS$ (vertically opposite angles)
$\therefore \Delta PRT \,|||\, \Delta QST$ (Two pairs of corresponding sides are in the same ratio and the included angles are equal.)

8—Using similar triangles

A 1 A–D, B–C, E–G, F–H 2 $\frac{AB}{MN} = \frac{BC}{LN} = \frac{AC}{LM}$

B 1 $x = 12$ cm 2 $x = 6$ cm, $y = 4$ cm
3 $x = 16$ cm, $y = 20$ cm 4 $y = 12$ cm
5 $x = 36$ cm, $y = 45$ cm 6 $x = 20$ cm, $y = 18$ cm
7 $x = 15$ cm, $y = 26$ cm 8 $x = 5.4$ cm, $y = 12.5$ cm
9 $x = 12$ cm 10 $x = 14.4$ cm, $y = 25$ cm

C 1 $x = 10$ cm 2 $x = 22.5$ cm 3 $x = 9$ cm
4 $x = 12.5$ cm, $y = 16.25$ cm 5 6 m
6 1.05 m or 105 cm

9—Similar triangles and reasoning

A 1 yes 2 yes 3 no 4 yes 5 no 6 yes

B 1 $\Delta ABE \,|||\, \Delta DCE$ (they are equiangular, $\angle A = \angle D$, $\angle B = \angle C$)
2 $\Delta ABE \,|||\, \Delta ACD$ ($\frac{AB}{AC} = \frac{AE}{AD} = \frac{5}{9}$ $\angle A$ is common)
3 $\Delta ADE \,|||\, \Delta CBE$ (matching sides are in the ratio 3:2)
4 b 30 cm
5 b 26 cm
6 b 6.4 cm
7 b 21 cm
8 b 5.25 cm
9 b 15 cm

C 4 b No, the pairs of alternate angles are *not* equal
8 b 1 : 2

10—Areas of similar figures

A 1 a 3 : 4 b A = 9 cm², B = 16 cm² c 9 : 16
2 a 1 : 2 b 1 : 2 c X = 12 cm², Y = 48 cm²
d 1 : 4
3 a 3 : 2 b 3 : 2 c A = 54 cm², B = 24 cm² d 9 : 4

B 1 a 1 : 2 b 1 : 4
2 a Corresponding sides are in the same ratio. b 1 : 2
c 1 : 4
3 9 : 25 4 a 3 : 4 b 9 : 16 5 64cm²
6 25.2 cm² 7 100 cm² 8 40 cm² 9 4
10 a 16 : 9 b 4 : 3 11 5 : 3 12 22.5 cm

C 1 a 1 : 2 b 16 cm², 96 cm² c 64 cm², 384 cm²
d 1 : 4 e 1 : 4 f A = 64 cm³, B = 512 cm³ g 1 : 8
2 a 1 : 3 b 108 cm² c 972 cm² d 1 : 9
e 72 cm³ f 1944 cm³ g 1 : 27
3 a If the lengths of corresponding edges are in the ratio a : b, then the surface areas are in the ratio $a^2 : b^2$.
b If the lengths of corresponding edges are in the ratio a : b then the volumes are in the ratio $a^3 : b^3$.